Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds 4 or more correct row minima/col maxima	AO1.1a	M1	Row minima = (2, 0, −2) Col maxima = (2, 3, 5)
	Correctly finds all row minima/col maxima	AO1.1b	A1	Max(row minima) = 2 Min(col maxima) = 2
	Explains correctly that as the max(row minima) is equal to the min(col maxima) then a stable solution exists and the value of the game is 2 Allow 'Maximin' and 'Minimax'	AO2.4	E1	As Max(row minima) = 2 = Min(col maxima), then a stable solution exists and the value of the game is 2
	Marking Instructions	**AO**	**Marks**	**Typical Solution**
ALT (a)	Uses dominance to reduce the size of the pay-off matrix to 2 × 3 or 3 × 2	AO1.1a	M1	$\mathbf{A}_1$ (or $\mathbf{A}_2$) dominates $\mathbf{A}_3$, so remove $\mathbf{A}_3$ [from pay-off matrix]
	Uses dominance to reduce the pay-off matrix to 1 × 1 OR Correctly finds all remaining row minima/col maxima	AO1.1b	A1	$\mathbf{S}_1$ [now] dominates $\mathbf{S}_2$ and $\mathbf{S}_3$, so remove $\mathbf{S}_2$ and $\mathbf{S}_3$ $\mathbf{A}_1$ [now] dominates $\mathbf{A}_2$, so remove $\mathbf{A}_2$
	Explains that each player will play the same strategy each time which results in Alex gaining 2 each game OR Explains correctly that as the max(row minima) is equal to the min(col maxima) then a stable solution exists and the value of the game is 2 Allow 'Maximin' and 'Minimax'	AO2.4	E1	Alex will only ever play $\mathbf{A}_1$ and Sam will only ever play $\mathbf{S}_1$, which results in Alex gaining 2 each game. Therefore, the value of the game is 2.
(b)	Identifies strategy for each player Condone stating '$\mathbf{A}_1$' and '$\mathbf{S}_1$'	AO1.1b	B1	Alex strategy $\mathbf{A}_1$ Sam strategy $\mathbf{S}_1$
				Total 4 marks

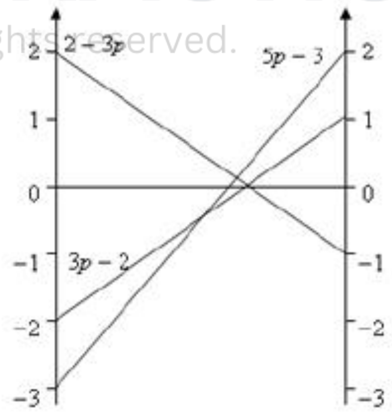
Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies correctly all three row minima or all three (two) column maxima	AO1.1a	M1	Row minima: −1, −2, (−1) Column maxima: 4, 0, (1)
	Determines correctly max(row min) and min(col max)	AO1.1b	A1	max (row min) = −1 min (col max) = 0
	Makes correct statement for the play-safe strategies of each player	AO3.2a	A1 CAO	Victoria plays *R* (or *P*); Albert plays *Y*
(b)	Identifies strategy and provides justification with reference to dominance	AO2.4	E1	Albert should never play *Z* because strategy *Y* dominates strategy *Z*

(c) (i)	States correctly that strategy *R* dominates strategy *P* (PI)	AO1.1a	B1	Victoria should never play *P* because strategy *R* dominates strategy *P*
	Introduces and defines a probability variable	AO3.3	M1	Let Victoria play strategy *Q* with probability p and strategy *R* with probability $1 - p$.
	Finds correctly both expected gain expressions for Victoria	AO1.1b	A1	If Albert plays: *X*: expected gain for Victoria $= -2p + 4(1 - p) = 4 - 6p$
	Clearly states that, as the game has no stable solution, the optimal solution occurs when the two expected gain expressions are set equal to each other, and then solves correctly 'their' equation constructed using 'their' expected gains for Victoria	AO2.2a	A1F	*Y*: expected gain for Victoria $= -(1 - p) = p - 1$ No stable solution, so optimal value of p occurs when: $4 - 6p = p - 1$, $p = 5/7$
	Interprets correctly the solution to the problem in the context, giving the optimal mixed strategy for Victoria	AO3.4	E1	Victoria should play strategy *Q* with a probability of 5/7 and *R* with a probability of 2/7
(ii)	Substitutes 'their' value of p into one of 'their' expected gain expressions, resulting in a	AO1.1b	B1F	4 – 6 × (5/7) = –2 / 7 or (5/7) –1 = – 2/7

	Marking Instructions	AO	Marks	Typical Solution
	value of the game			
(iii)	Recognises correctly that Victoria can improve on the value of the game if Albert does not play an optimal mixed strategy	AO3.5b	E1	The expected pay-off for Victoria will only be −2/7 if Albert plays an optimal mixed strategy between *X* and *Y*
				Total 11 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
Deduces strategy *C* is a dominated strategy	AO2.2a	B1	−3 = −3, −4 < −2, 1 < 2, hence strategy *B* dominates strategy *C*
Introduces and defines a probability variable	AO3.3	M1	Let John choose strategy *A* with probability p and strategy *B* with probability $1 - p$. If Danielle plays:
Finds correctly all three expected gain expressions for John	AO1.1b	A1	*X*: expected gain for John = $2p - 3(1 - p) = 5p - 3$ *Y*: expected gain for John = $p - 2(1 - p) = 3p - 2$
Constructs a graph with at least one vertical axis and plots one of 'their' expected gains correctly (PI)	AO1.1a	M1	*Z*: expected gain for John = $-p + 2(1 - p) = 2 - 3p$
Identifies correctly the optimal point of intersection from 'their' graph and finds 'their' value of probability variable	AO1.1b	A1	
Interprets correctly the solution to the problem in the context, giving the optimal mixed strategy for John	AO3.2a	E1	$3p - 2 = 2 - 3p$ $6p = 4$ $p = 2/3$ John should play *A* with probability 2/3

			and *B* with probability 1/3 (and *C* with probability 0)
			Total 6 marks

Q4.

(a) Min

$$\begin{array}{c} \\ \\ \\ \text{Max} \end{array}\begin{array}{c}\begin{pmatrix} 4 & -1 & 2 & 3 \\ 4 & 6 & 3 & 7 \\ 1 & 3 & -2 & 4 \end{pmatrix} \\ \begin{matrix} 4 & 6 & 3 & 7 \end{matrix}\end{array}\begin{array}{c} -1 \\ 3 \\ -2 \\ \\ \end{array}$$

Maximin (row) = 3

Either correct, including correct values

M1

Minimax (col) = 3

Both correct, written as equations PI by next line

A1

CSO

As Maximin (row) = Minimax (col)
There is a stable solution

Must have equation and statement and scored first 2 marks

E1

(Play safe) (H) B
(Play safe) (W) F

Both correct

B1

4

(b) Saddle point (B, F)

B1

1

[5]

Q5.

(a) R min −4, −5, −2 plays C

Either C or E stated

B1

J max 4, 1, 3 plays E

Both C and E stated

B1

and all values shown

E1

3

(b) maximin R = −2 ≠ 1 = minimax J

Correct values must be stated

E1

1

(c) (For Juliet,) col E dominates col D

E1

1

(d) (i) Signs changed as J gains = R losses

E1

Gains written as rows

E1

2

(ii) Let J play E prob p

F $(1 - p)$

If R plays A, J wins $4p$

B $5p - 3(1 - p)$

2 correct expressions seen

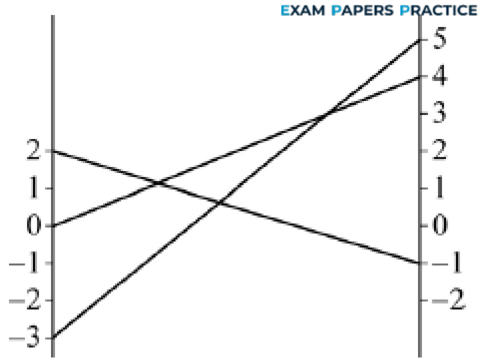
M1

C $-p + 2(1 - p)$

All correct

A1

[gives $4p$, $8p - 3$, $2 - 3p$]



Must have 3 lines

m1

All correct with values shown

A1

Max at $8p - 3 = 2 - 3p$

Identifies correct max from their graph

m1

$p = \frac{5}{11}$

A1

(J plays)E prob $\frac{5}{11}$, F prob $\frac{6}{11}$

A1

CSO

7

(iii) Value of game = $\frac{7}{11}$

B1

1

[15]

Q6.

(a) (i)

Row min − 6, − 3, − 5, − 4

Max (row min) = − 3

Col max 5, 4, − 3

Min (col max) = − 3

attempt to find maximin and minimax
condone one slip in values

M1

all** rows min and col max **values correct
and** max (row min) = − 3 **identified
and** min (col max) = − 3 **identified

A1

max (row min) = min (col max) = − 3
⇒ game has a stable solution

***full** statement involving maximin and minimax **and** both values = − 3*

E1

3

(ii) Adam plays $\mathbf{A_2}$ & Bill plays $\mathbf{B_3}$

B1

1

(iii) Value of game for Bill is +3

B1

1

Examiners must use the correct symbol for marks carried forward, ie ringed totals with arrows through them.

(b) (i) (Never play) $\mathbf{C_2}$

correct strategy stated

C_2 dominated **by** C_1 (–3 > –4 and 2 > 1)

***and** correct reason*
condone $3 < 4$ and $-2 < -1$

B1

1

(ii) C_1 : $3p - 2(1 - p)$

either correct unsimplified

M1

C_3 : $-3p + 5(1 - p)$

both correct unsimplified
{$5p - 2$, $5 - 8p$}

A1

2

(iii) $3p - 2(1 - p) = -3p + 5(1 - p)$

equating their 2 expressions

M1

$\Rightarrow \quad p = \frac{7}{13}$

A1

2

(iv) Value of game $= 5 \times \frac{7}{13} - 2$

or $5 - 8 \times \frac{7}{13}$

$= \frac{9}{13}$

B1

1

[11]

Q7.

(a) Row minima are –4, –3, –7

__both__ row minima __and__ column maxima attempted (condone 2 errors)

M1

Column maxima are –3, 6, 8

all values correct

A1

max (row min) = min (col max) = −3

condone arrows pointing to this element but must __state__ max (row min) and min (col max) or equivalent

E1

Play-safe Tom **II** and Jerry **A**

B1

4

(b) (i) Let Rohan play R_1 with prob p

$\Rightarrow$ plays R_2 with prob $1 - p$

When Carla plays C_1 ,

Rohan's expected gain $= 3p + (1 - p)$

$= 1 + 2p$

$C_2 : 5p + (-2)(1 - p) = 7p - 2$

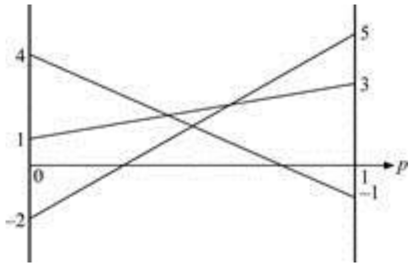
at least 2 expected gains correct unsimplified

M1

$C_3 : -p + 4(1 - p) = 4 - 5p$

all 3 correct unsimplified

A1



at least 2 lines correct

M1

all lines correct for $0 \le p \le 1$ and values at 0 and 1 clear

A1

$7p - 2 = 4 - 5p$

$12p = 6$

choosing highest point or using correct equation

M1

$\Rightarrow p = \frac{1}{2} \Rightarrow$ Rohan plays R_1 50% of the time and R_2 50% of the time

A1cso

Value of game $= 7 \times \frac{1}{2} - 2 = \frac{3}{2}$ AG

or $4 - \frac{5}{2} = \frac{3}{2}$ *must see working*

B1

7

(ii) When Rohan plays R_1, expected loss for Carla is $3p + 5q + (-1)(1 - p - q)$

and when Rohan plays R_2, expected loss for Carla is $p + (-2)q + 4(1 - p - q)$

either expression correct unsimplified

M1

$4p + 6q = \frac{3}{2} + 1$

$3p + 6q = 4 - \frac{3}{2}$

correct simultaneous equations unsimplified

A1

$\Rightarrow p = 0,\ q = \frac{5}{12}$

condone 0.42 or better

A1

$\Rightarrow$ Carla never plays C_1,

plays C_2 with prob $\frac{5}{12}$

and plays C_3 with prob $\frac{7}{12}$

Must have all 3 correct probabilities

E1cso

4

[15]

Q8.

(a) (i) Roseanne plays R_1 with prob p

Expected value when Collette plays

$C_1 : -3p + 2(1 - p) = 2 - 5p$

$C_2 : 2p - (1 - p) = 3p - 1$

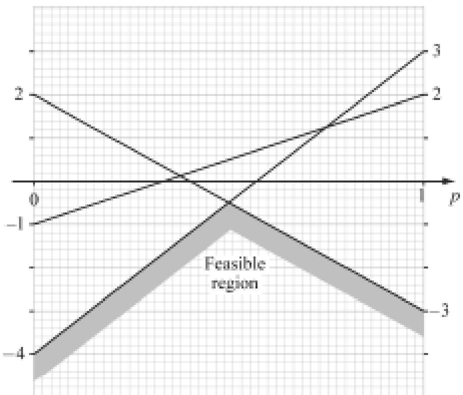
$C_3 : 3p - 4(1 - p) = 7p - 4$

One correct unsimplified

M1

All correct unsimplified

A1



drawing 'their' lines (2 'correct' ft)

M1

correct with values clear at p = 0 and p = 1

A1

Solving $2 - 5p = 7p - 4$

$6 = 12p$

their highest point

M1

$\Rightarrow p = \frac{1}{2}$

A1

SC B1 if p = ½ found from graph

Strategy is to play R_1 for 50% of time

E1ft

7

(ii) Value $= 2 - 5\left(\frac{1}{2}\right)$ or $7\left(\frac{1}{2}\right) - 4 = -\frac{1}{2}$

AG CSO

$p = \frac{1}{2}$ *and both expressions correct*

B1

1

(b) (i) Let Collette play C_1 with prob p

and C_2 with prob q

$\Rightarrow C_3$ with prob $1 - p - q$

B1

1

(ii) $-3p + 2q + 3(1 - p - q) = -\frac{1}{2}$

$2p - q - 4(1 - p - q) = -\frac{1}{2}$

Either equation LHS correct

Condone (1 – p + q) used

M1

$\Rightarrow 6p + q = 3\frac{1}{2}$

$6p + 3q = 3\frac{1}{2}$

Either equation

correct and simplified p & q coefficients

A1

$\Rightarrow p = \frac{7}{12}$

$q = 0$

CSO

A1

$\Rightarrow$ Collette plays C_1 with prob $\frac{7}{12}$,

(never plays C_2),

and plays C_3 with prob $\frac{5}{12}$

Must have statement with C_1 & C_3
correct only

E1 4

[13]

Q9.

(a) (i) Min R_1 (5, 2, –1) = –1
Min R_2 (–3, –1, 5) = –3
Min R_3 (4, 1, –2) = –2

E1

Max min = –1
$\Rightarrow$ Play safe strategy R_1

B1 2

(ii) Max C_1 = 5; max C_2 = 2; max C_3 = 5
Min (5, 2, 5) = 2

M1

$2 \neq 1 \Rightarrow$ no stable solution

A1 2

(b) R_3 (4, 1, –2) < R_1 (5, 2, –1)

E1 1

(c) (i) C_1 played, expected gain for Rose:
$5p + -3(1 - p)$

Any correct expected gain unsimplified

M1

$= 8p - 3$

One correct simplified

A1

$C_2 \div p - (1 - p) = 3p - 1$

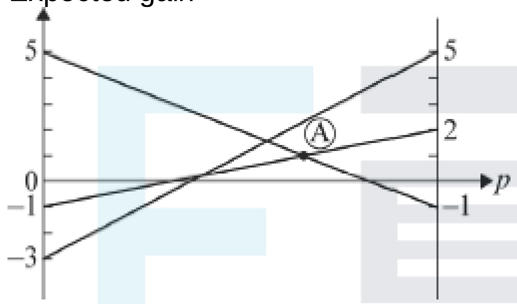
$C_3 \div -p + 5(1 - p) = 5 - 6p$

All correct simplified

A1

3

(ii) Expected gain



Plotting at least 2 lines

M1

All correct with values at p = 0 and p = 1 indicated

A1

2

(iii) Choosing A – highest point in feasible region

$\Rightarrow 3p - 1 = 5 - 6p$

$9p = 6$

Solving this equation

M1

$\Rightarrow p = \frac{2}{3}$

CSO

A1

$\Rightarrow$ Rose plays R_1 $\frac{2}{3}$ of time

and R_2 $\frac{1}{3}$ of time

E1ft

3

(iv) Value of game = $3 \times \frac{2}{3} - 1 = 1$

Or 5 – 4 = 1

B1

1

[14]

Q10.

(a) (–2,2,4) < (2,4,5)

So S_1 dominated by S_2

E1

$$\begin{pmatrix}4\\5\\2\end{pmatrix} > \begin{pmatrix}2\\4\\1\end{pmatrix}$$

note > sign

So $_3$ dominated by C_2

E1

2

(b) C_1 C_2

2 × 2 game now $\begin{array}{c c} S_2 & \\ S_3 & \end{array}\begin{bmatrix}2 & 4\\5 & 1\end{bmatrix}$

Minimum of rows (2, 4) = 2

correct method for either S or C

M1

Minimum of (5, 1) = 1

Choose maximum = **2**

play safe for Sam is S_2

A1

Max of column 1 = max (2,5) = 5

Max of column 2= max (4,1) = 4

Choose minimum = 4

play safe for computer is C_2

AI

Since 2 ≠ 4 $\Rightarrow$ not stable solution

E1

4

(c) (i) Computer picks C_1

Expected game $= 2p + 5(1 - p)$

M1

$= 5 - 3p$

A1

Computer picks C_2

Expected gain $= 4p + (l - p)$

$= 1 + 3p$

A1

3

(ii) Best mixed strategy

$5 - 3p = 1 + 3p$

M1

$\Rightarrow p = \frac{2}{3}$

M1 A1

2

(iii) Expected points gain

$= 5 - 3 \times \left(\frac{2}{3}\right)$

Or $1 + 3\left(\frac{2}{3}\right)$

$= 3$

B1

1

[12]

Q11.

(a) Gain for Rowan +gain for Colleen in each strategy = 0

Gain for one = loss of other

E1

1

(b)

	–3	–4	1	–4
	1	5	–1	$\boxed{-1}$
	–2	–3	4	–3
Max	$\boxed{1}$	5	4	

$$\begin{cases} \textit{minimum of rows \& max of columns} \\ \textit{or} \\ \textit{maximum of minima or minimax} \end{cases}$$

M1

All values correct (seen) or words maximin and minimax highlighted

A1

$1 \neq -1 \Rightarrow$ no stable solution

E1 **3**

(c) R_3 dominates R_1
$(-3, -4, 1) < (-2, -3, 4)$ so never play R_1

E1 **1**

(d) (i) R chooses R_2 with prob p

Attempt at one expression

M1

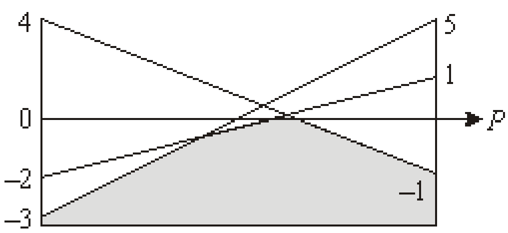
$\Rightarrow$ choose R_3 with prob $1 - p$
$\Rightarrow$ expected gain when C plays
$C_1 : p - 2(1 - p) = 3p - 2$
$C_2 : 5p - 3(1 - p) = 8p - 3$
$C_3 : - p + 4(1 - p) = 4 - 5p$

All correct unsimplified

A1

Plot expected gains for $0 \le p \le 1$

M1



Condone mirror image

A1

Choosing their "highest" point

Any 2 lines

M1

C_1 & C_3 intersect $\Rightarrow$ $3p - 2 = 4 - 5p$

$\Rightarrow p = \frac{3}{4}$

A1

$\left.\begin{array}{l}\Rightarrow \text{play } R_2 \text{ with prob } \frac{3}{4} \\ \text{and } R_3 \text{ with prob } \frac{1}{4}\end{array}\right\}$

Statement of strategy

E1ft

7

(ii) Value of game is $3 \times \frac{3}{4} - 2 = \frac{1}{4}$

CSO or equivalent, e.g. 0.25

B1

1

[13]

Q12.

(a) (i) Row min (–2,2 –3)

M1

Max 2

Column max (3, 5, 2, 6)

Min 2

A1

2 = 2, therefore stable solution

E1

3

(ii) 1 saddle point at (2, 3)

B1

1

(b) Colin plays I with prob p, II with $(1 - p)$

M1

Return $p(x + 2) + 3(1 - p)$

A1

$p(x - 1) + 5(1 - p)$

A1

$p(x - 1) + 3 = \frac{19}{5}$

equating to value

M1

$p(x - 6) + 5 = \frac{19}{5}$

$p(x - 1) = \frac{4}{5}$

or solving for p first

A1

$p(x - 6) = -\frac{6}{5}$

A1

$$\frac{4}{5(x-1)} = \frac{-6}{5(x-6)}$$

M1

$4x - 24 = -6x + 6$

$x = 3$

CAO

A1

8

[12]

Q13.

(a) Min rows $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ Max = 2

M1

Max cols Min = 4

(7, 4, 6)

A1

$2 \neq 4$

E1

3

(b) III > II

B1

$$\begin{bmatrix} 1 & 3 & 6 \\ 7 & 4 & 2 \end{bmatrix}$$

B1

2

(c) Let Bev play with p, q, $1 - p - q$

M1

$p + 3q + 6(1 - p - q) = 3\frac{3}{5}$

A1 without $\left(= 3\frac{3}{5}\right)$

A1

$7p + 4q + 2(1 - p - q)\ 3\frac{3}{5}$

A1

$-5p - 3q = -2\frac{2}{5}$

$5p + 2q = 1\frac{3}{5}$

$\Rightarrow q = \frac{4}{5}$, $p = 0$

A1

Plays I prob 0

II prob $\frac{4}{5}$

III prob $\frac{1}{5}$

All three

B1

5

[10]

Q14.

(a) Not drop shot
as R1 > R2
or R3 > R2

B1

E1

2

(b) D chooses B p
D chooses L $1 - p$

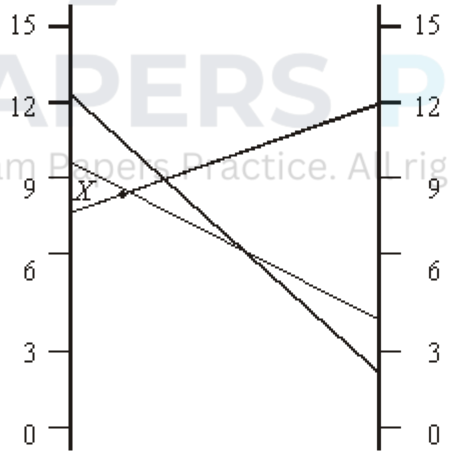
M1

If T chooses B $\Rightarrow$ $12p + 8(1 - p)$
D $\Rightarrow$ $5p + 10(1 - p)$
L $\Rightarrow$ $2p + 13(1 - p)$

A1

B = $4p + 8$ (8, 12)
D = $10 - 5p$ (10, 5)
L = $13 - 11p$ (13, 2)

A1



M1 A2

Max at X

B1

$4p + 8 = 10 - 5p$

$p = \frac{2}{9}$

B1

David should play
Boast 2 shots to every Lob 7 shots (2 : 7)
Both

B1 8

(c) Value = $\frac{8}{9}$ + 8
= $8\frac{8}{9}$

B1F 1

[11]

Q15.
(a) 7

B1 1

(b) Col. Max 8, 13, 8, 11

M1

Min = 8

A1

Row Min –7, –6, 8, 8
Max = 8

A1 3

∴ Stable solution

(c) Points (3,1) (4,1) (3,3) (4,3)

B2,1 2

[6]

Examiner reports

Q1.

Nearly 90% of students scored at least one mark in part (a). Of these students, many scored 2 marks but failed to score the third mark due to a lack of depth in their explanation of why the value of the game was 2. Many did not refer to, or were not clear enough about, the max(row minima) and min(col maxima) both being equal to 2 and therefore being equal to each other, and many did not then relate this back to the value of the game.

Interesting correct solutions included the use of dominance to reduce the size of the pay-off matrix to 1 × 1. Of the errors that students made, the most prevalent was determining the row maxima and column minima, which lead to an incorrect conclusion and often impacted the answer to (b).

Q4.

Although the majority of students had an idea of the requirements of this question, full marks were rarely achieved. It is essential that students give full and clear statements to provide evidence to the examiner; simply writing down seven numbers, circling two of the numbers, then stating '3 = 3' will not score full marks. Students needed to show the seven numbers, but then they had to state; 'maximin(row) = 3' and 'minimax(col) = 3', followed by 'as maximin(row) = minimax(col), there is a stable solution'. The play-safe strategies were well-answered, but there were very few correct answers for the saddle point.

Q5.

In parts (a) and (b), the majority of students had an idea of the requirements of this question, but full marks were rare. It is essential that students give full and clear statements to convince an examiner; simply writing down 6 numbers, circling two of the numbers, then stating '1 ≠ – 2' will not score full marks. Part (c) was well answered, although some students lost the mark through a lack of clarity in their statement. Part (d)(i) was poorly answered. Although students realized that the question was concerned with changing signs, the explanations given were poor and showed a lack of understanding of the problem. Many students scored full marks on the final 2 parts. Errors occurred when students failed to draw accurate graphs and consequently failed to identify the correct maximum point.

Q6.

(a) (i) Explanations rarely scored full marks. In order to show that the game has a stable solution, it was expected that the maximum values in the columns would be indicated before finding the minimum value of these maxima. The maximum of the row minima also needed to be found. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. Quite a few students felt that all they needed to do was to draw arrows pointing to –3 when more detail was required. A few chose to solve the problem using an argument based on dominance, but the explanations were rarely adequate to earn more than a single method mark.

(ii) It was encouraging to see many students correctly identify the play-safe strategy for each player.

(iii) Only the better students stated the correct value of the game for **Bill**, despite the bold type; as you might expect weaker students gave the value as – 3, and others gave the value as 0, possibly confused by the term “zero-sum game”.

(b) *A large number of students were not prepared for this part of the question*.

(i) A large number deleted the first column and scored only a couple of method marks for the rest of this question. Some who realised that the computer should not play C_2 spoiled their answers by writing such things as “each of the other strategies is better”.

(ii) The better students managed to use the values in the table correctly to find correct expressions for the expected gains for Roza for the two remaining strategies. A few ignored the dominance from part(b)(i) and attempted to solve the problem graphically using three lines; some successfully, others not.

(iii) Some poor algebra, usually when removing brackets, prevented some from finding the correct value of p, but most marks were lost for having the incorrect pair of simultaneous equations.

(iv) Those who scored full marks in parts (b)(ii) and (iii) were usually able to obtain the correct value of the game for Roza.

Q7.

(a) Explanations rarely scored full marks. In order to show that the game has a stable solution, it was expected that the maximum values in the columns would be indicated before finding the minimum value of these maxima. The maximum of the row minima also needed to be found. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. Quite a few candidates felt that all they needed to do was to draw arrows pointing to –3 when an explicit statement was required regarding the play-safe strategies for each player.

(b) (i) Most candidates were able to find the optimal mixed strategy for Rohan. Very few made a statement such as “Let Rohan play R_1 with probability p”; usually the letter p was introduced with no explanation. Some lost marks for not drawing accurate graphs of expected values for $0 \leq p \leq 1$. Although it is acceptable to use the horizontal lines printed in the answer book, at least one of these vertical lines should have a clear scale so that the accuracy of the process of finding the highest point in the region can be checked by the examiner. Those finding the correct optimal strategy were usually able to find the correct value of the game, but for full marks it was necessary to make a statement regarding the frequencies of playing the two strategies.

(ii) A large number of candidates were not prepared for this part of the question. Despite the wording in the question, very few made the correct start using the **three** probabilities p, q and $1 - p - q$ in order to obtain a pair of simultaneous equations in p and q. Those who made a correct start usually completed and found the optimal mixed strategy for Carla; some candidates, however, did obtain values for p and q which were either negative or more than 1.

Q8.

Despite the standard nature of part (a)(i), it was surprising to see many candidates unaware of how to find the optimal mixed strategy. A good sketch showing the feasible region was expected with the highest point of the feasible region being selected in order to find the probability of playing the various rows. Many candidates gave no reason for introducing a probability p nor explained what the actual mixed strategy was when the value of p had been found.

In part (a)(ii), it was quite surprising to see a number of candidates substituting $p = 0.5$ into their third (unused) expression and consequently obtaining the wrong value of the game, even though the answer was printed in the question.

In part (b)(i), almost everyone obtained the correct answer of $1 - p - q$. Part (b)(ii) was more discriminating, and it was obvious that some were totally unfamiliar with the technique needed. Many did not see the need to use the value of the game, and others made errors in their algebra. Nevertheless, there were quite a few completely correct solutions to this part of the question.

Q9.

Part (a)(i) proved to be the most testing part of this question, with very few candidates giving a good reason why Rose should choose R_1 as the play-safe strategy. The minimum value in each row had to be calculated and then the maximum of these minima needed to be shown to equal –1. In part (a)(ii), the better solutions gave a table with the column maxima 5, 2 and 5 clearly stated and with the minimum of these values indicated. A statement that the minimax value, 2, was not equal to the maximin value, –1, from part (a)(i) was then required to show that the game has no stable solution.

In part (b), most candidates realised that R_1 dominates R_3 and showed that corresponding entries of R_1 were greater than those of R_3.

The expressions for the expected gains in part (c) were usually correct, although some made algebraic slips when trying to simplify their expressions. There was a casual approach by many when drawing the graphs. Lines were often too short or too long and many made no attempt to indicate values on the vertical axis. Candidates do not appear to know what the three lines on the graph represent. Some seemed to guess which two expressions to equate when finding the optimal mixed strategy for Rose, instead of considering the highest point in the feasible region.

It was necessary to make a statement indicating that Rose plays R_1 for 2/3 of the time and R_2 for 1/3 of the time in order to score full marks in part (c)(iii). It was quite surprising to see a number of candidates substituting $p = 2/3$ into their third (unused) expression and consequently obtaining the wrong value of the game.

Q10.

The idea of dominance seems to be well understood. Marks were lost in part (b) for not identifying the actual play safe strategies. It is not sufficient to draw circles around a couple of numbers and to simply state there is no saddle point. Unfortunately, this is precisely what many candidates offered as their solution. This question was a simple example of game theory, reducing to a 2 by 2 game, and most candidates wrote down and solved correctly an equation in p and hence the expected points gain for Sam. Consequently, it was pleasing to see many high marks in this question.

Q11.

The idea of dominance seems to be well understood but only the best candidates could explain the term ‘zero-sum game’. In order to show that the game has no stable solution, it is expected that the minimum values in the rows and maximum values in the columns be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then be made indicating that these two values are not equal and hence the game has no stable solution. To find the optimal mixed strategy, when a letter such as p is introduced, there should be an indication that this is the probability that Rowan is choosing R_2 for example. Three expressions in /? were often written down with no indication as to what they represented. When three linear graphs had been drawn, many candidates made mistakes in choosing the appropriate highest point of the feasible region. Again, this gave the impression that many were following a recipe rather than understanding what this selection meant. Those who obtained the correct optimal strategy were able to find the value of the game.

Q12.

Again this question was well answered, with many complete solutions to a conceptually difficult question. This was the first time that a game theory question had been set in which the values were algebraic. Some candidates solved for p first, others for x, with equal success.

Q13.

Candidates knew the theory for part (a) of this question but there were very few scripts that earned full marks. Candidates had to show calculations to answer this question rather than simply stating that there is no play safe solution.

Part (b) was well answered.

Part (c) was the least well answered question on the paper. Some candidates adopted the correct approach, using probabilities p, q, and $1 - p - q$, but still found the completion of the question difficult. There were, however, a number of correct solutions.

Q14.

There were some excellent answers to this question, with candidates showing a clear understanding of game theory. However, some candidates were less well prepared. Despite drawing correct diagrams, they were unable to identify the correct solution to the problem, and a value of 0.5 for p was frequently obtained.

Q15.

This question discriminated well. Candidates who had a good understanding of game theory scored maximum marks but the majority of candidates could not correctly identify the saddle points. It was essential that candidates were able to analyse all the different aspects of game theory mentioned in the specification.